

Node.js Backend Developer Curriculum

1. Introduction to Internet & Backend
 - a. What is Backend Development?
 - b. Why is the Backend Needed?
 - c. What is a Server?
 - d. Client-Server Architecture
 - e. 3-Tier Architecture (Presentation, Application, Database layers)
 - f. Request-Response Lifecycle
 - g. How the Internet Works (DNS, IP, Ports, Protocols)
 - h. HTTP vs HTTPS
2. Introduction to Node.js
 - a. What is Node.js?
 - b. Why Node.js?
 - c. History of Node.js
 - d. Applications of Node.js
 - e. Installing Node.js and npm
 - f. Using REPL (Read-Eval-Print Loop)
 - g. Using CLI to run files
 - h. Writing a "Hello World" Node.js program
 - i. Creating simple node js server using http module
3. Modules in Node js and custom modules
 - a. Introduction to Modules
 - b. Types of Modules in Node.js
 - c. Creating Custom Modules (Default and Named Exporting)
 - d. Custom modules using import and require
 - e. Local (User-defined) Modules – CommonJS vs ES Modules
4. Built-in Modules
 - a. Introduction to Built-in Modules: fs, path, os, http
 - b. FS module and path module
 - i. fs and fs/promises modules
 - ii. Path module, path module with fs module
 - iii. fs module methods(Sync vs Async)
 - iv. Streams and Buffers
 - c. OS module, HTTP module
5. Third party modules
 - a. Installing and using npm modules like nodemon, chalk, colour, day js, [express.js](#), multer, node mailer, etc

6. Node Architecture
 - a. Event Loop
 - b. Non-blocking I/O
 - c. Single-threaded model
7. Building APIs with Node.js
 - a. What is an API?
 - b. SOAP vs REST APIs
 - c. Creating a Server with http and url modules
 - d. Handling Query Params and Request Body
 - e. Building Basic CRUD APIs (GET, POST, PUT, PATCH, DELETE)
 - f. Testing APIs with Postman
8. Introduction to Express.js
 - a. What is Express.js and Why Use It?
 - b. Installing Express
 - c. Creating a Basic Express Server
 - d. Handling GET, POST, PUT, DELETE Requests
 - e. Using req and res Objects
 - f. Creating a Basic Express API with Multiple Routes
9. Deployment of API using render.com and CORS
 - a. Deployment of APIS
 - b. What is CORS and How to Use cors Module
10. Middleware
 - a. What is Middleware?
 - b. Built-in Middleware (express.json(), express.static())
 - c. Creating Custom Middleware
 - d. Router-level Middleware with express.Router()
11. Error Handling and MVC Architecture
 - a. Error-handling Middleware
 - b. Introduction to MVC Architecture
 - c. Structuring Express Applications (Controllers, Routes, Models)
 - d. Mini Project: Simple E-commerce API using Express and File System
12. Introduction to MongoDB
 - a. What is NoSQL and Why MongoDB?
 - b. Database, Collections, Documents
 - c. Installing MongoDB
 - d. MongoDB CRUD Operations
 - e. MongoDB Operators
 - f. MongoDB Aggregations

13. MongoDB & Mongoose CRUD Operations

- a. Integrating MongoDB using Mongoose with Nodejs/Express
- b. Environment Variables with dotenv
- c. Creating Mongoose Models and Schema using MVC Architecture
- d. Performing CRUD Operations (Create, Read, Update, Delete)
- e. Query Operators and Filtering
- f. Hands-on: Build CRUD API using Express and Mongoose

14. Authentication & Authorization

- a. Introduction to Authentication & Authorization
- b. Password Hashing with bcryptjs
- c. Sending Emails using nodemailer after account creation
- d. JWT (JSON Web Tokens) for Authentication
- e. Role-Based Access Control (Admin/User roles)
- f. Protecting Routes with Middleware
- g. crud operation with apis using Referencing and Populating Documents (ref, populate)

15. File Uploads & Validation

- a. Handling File Uploads with multer
- b. Uploading to Cloudinary
- c. Input Validation using express-validator
- d. Hands-on: Create an API endpoint for file upload and validation

16. Deployment on Render

- a. Preparing Your App for Production
- b. Environment Configuration for Production
- c. Connecting MongoDB Atlas
- d. Deploying Node.js + MongoDB on Render (Free Hosting)
- e. Testing Deployed API with Postman

17. Capstone project (3 days)

Day wise topics:  Node daywise topics